

Naïve Bayes: Spam Filter Worked Example

Training Data

We use a toy vocabulary of 5 words: {free, click, meeting, report, offer}, and $N = 6$ labelled training emails ($x_j = 1$ if word j appears, 0 otherwise). There are $K = 2$ classes: spam ($k = 1$) and ham ($k = 0$).

Email	free	click	meeting	report	offer	Label
1	1	1	0	0	1	Spam ($y = 1$)
2	1	0	0	0	1	Spam ($y = 1$)
3	0	1	0	0	0	Spam ($y = 1$)
4	0	0	1	1	0	Not spam ($y = 0$)
5	0	0	1	0	0	Not spam ($y = 0$)
6	0	0	0	1	0	Not spam ($y = 0$)

There are **3 spam** emails ($y = 1$, i.e. $k = 1$) and **3 legitimate** (ham) emails ($y = 0$, i.e. $k = 0$).

Raw MLE Estimates

The MLE for each parameter is the natural frequency count:

$$\hat{\phi}_{j|k} = \frac{\sum_{i=1}^N \mathbf{1}[x_j^{(i)} = 1 \wedge y^{(i)} = k]}{\sum_{i=1}^N \mathbf{1}[y^{(i)} = k]}, \quad \hat{\phi}_y = \frac{\sum_{i=1}^N \mathbf{1}[y^{(i)} = 1]}{N}.$$

Prior

$$\hat{\phi}_y = \frac{3}{6} = 0.5$$

Spam class likelihoods ($y = 1$) – denominator: 3

$$\begin{aligned} \hat{\phi}_{\text{free}|1} &= \frac{2}{3} \approx 0.667 && \text{(emails 1, 2 contain 'free')} \\ \hat{\phi}_{\text{click}|1} &= \frac{2}{3} \approx 0.667 && \text{(emails 1, 3 contain 'click')} \\ \hat{\phi}_{\text{meeting}|1} &= \frac{0}{3} = \mathbf{0.000} && \text{'meeting' never in spam} \star \\ \hat{\phi}_{\text{report}|1} &= \frac{0}{3} = \mathbf{0.000} && \text{'report' never in spam} \star \\ \hat{\phi}_{\text{offer}|1} &= \frac{2}{3} \approx 0.667 && \text{(emails 1, 2 contain 'offer')} \end{aligned}$$

Ham class likelihoods ($y = 0$) – denominator: 3

$$\begin{aligned} \hat{\phi}_{\text{free}|0} &= \frac{0}{3} = \mathbf{0.000} && \text{'free' never in ham} \star \\ \hat{\phi}_{\text{click}|0} &= \frac{0}{3} = \mathbf{0.000} && \text{'click' never in ham} \star \\ \hat{\phi}_{\text{meeting}|0} &= \frac{2}{3} \approx 0.667 && \text{(emails 4, 5 contain 'meeting')} \\ \hat{\phi}_{\text{report}|0} &= \frac{2}{3} \approx 0.667 && \text{(emails 4, 6 contain 'report')} \\ \hat{\phi}_{\text{offer}|0} &= \frac{0}{3} = \mathbf{0.000} && \text{'offer' never in ham} \star \end{aligned}$$

Prediction 1 (MLE)

Consider the email

$$x^{(A)} = (\text{free} = 1, \text{click} = 0, \text{meeting} = 0, \text{report} = 0, \text{offer} = 1).$$

Recall that $P(x_j = 0 \mid y = k) = 1 - \hat{\phi}_{j|k}$.

Spam score

$$\begin{aligned} &P(x^{(A)} \mid y = 1) \cdot P(y = 1) \\ &= \hat{\phi}_{\text{free}|1} \cdot (1 - \hat{\phi}_{\text{click}|1}) \cdot (1 - \hat{\phi}_{\text{meeting}|1}) \cdot (1 - \hat{\phi}_{\text{report}|1}) \cdot \hat{\phi}_{\text{offer}|1} \cdot \hat{\phi}_y \\ &= \frac{2}{3} \cdot \frac{1}{3} \cdot (1 - 0) \cdot (1 - 0) \cdot \frac{2}{3} \cdot 0.5 = \frac{2}{3} \cdot \frac{1}{3} \cdot 1 \cdot 1 \cdot \frac{2}{3} \cdot 0.5 = \frac{4}{9} \times 0.5 = \mathbf{0.222} \end{aligned}$$

Ham score

$$\begin{aligned} &P(x^{(A)} \mid y = 0) \cdot P(y = 0) \\ &= \hat{\phi}_{\text{free}|0} \cdot (1 - \hat{\phi}_{\text{click}|0}) \cdot (1 - \hat{\phi}_{\text{meeting}|0}) \cdot (1 - \hat{\phi}_{\text{report}|0}) \cdot \hat{\phi}_{\text{offer}|0} \cdot (1 - \hat{\phi}_y) \\ &= 0 \cdot 1 \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot 0 \cdot 0.5 = \mathbf{0} \end{aligned}$$

Prediction for $x^{(A)}$ (MLE)

Since $0.222 > 0$, we predict $\hat{y} = 1$ (spam). ✓

Prediction 2 (MLE): A Case That Breaks

Now consider an email that mixes words from both classes – entirely realistic in practice. A work email could advertise a “free” lunch at an upcoming “meeting”, or a phishing email could mimic a legitimate report.

$$x^{(B)} = (\text{free} = 1, \text{click} = 0, \text{meeting} = 1, \text{report} = 0, \text{offer} = 1).$$

This email contains the spam indicators **free** and **offer**, and also the ham indicator **meeting**.

Spam score

$$\begin{aligned} & P(x^{(B)} \mid y = 1) \cdot P(y = 1) \\ &= \hat{\phi}_{\text{free}|1} \cdot (1 - \hat{\phi}_{\text{click}|1}) \cdot \hat{\phi}_{\text{meeting}|1} \cdot (1 - \hat{\phi}_{\text{report}|1}) \cdot \hat{\phi}_{\text{offer}|1} \cdot \hat{\phi}_y \\ &= \frac{2}{3} \cdot \frac{1}{3} \cdot \underbrace{0}_{\hat{\phi}_{\text{meeting}|1}=0} \cdot 1 \cdot \frac{2}{3} \cdot 0.5 = 0 \end{aligned}$$

Ham score

$$\begin{aligned} & P(x^{(B)} \mid y = 0) \cdot P(y = 0) \\ &= \hat{\phi}_{\text{free}|0} \cdot (1 - \hat{\phi}_{\text{click}|0}) \cdot \hat{\phi}_{\text{meeting}|0} \cdot (1 - \hat{\phi}_{\text{report}|0}) \cdot \hat{\phi}_{\text{offer}|0} \cdot (1 - \hat{\phi}_y) \\ &= \underbrace{0}_{\hat{\phi}_{\text{free}|0}=0} \cdot 1 \cdot \frac{2}{3} \cdot \frac{1}{3} \cdot \underbrace{0}_{\hat{\phi}_{\text{offer}|0}=0} \cdot 0.5 = 0 \end{aligned}$$

Prediction for $x^{(B)}$ (MLE) – Classifier breaks down

Both scores are exactly zero:

$$P(x^{(B)} \mid y = 1) \cdot P(y = 1) = 0 \quad \text{and} \quad P(x^{(B)} \mid y = 0) \cdot P(y = 0) = 0.$$

The scores are tied at zero and no prediction can be made.

The spam score is killed by $\hat{\phi}_{\text{meeting}|1} = 0$: the model assigns zero probability to any spam email containing “meeting”, so the entire product collapses, regardless of the strong spam evidence from **free** and **offer**. Simultaneously, the ham score is killed by $\hat{\phi}_{\text{free}|0} = \hat{\phi}_{\text{offer}|0} = 0$. A single unseen word zeroes out the entire likelihood – this is the **zero-probability problem**.

The root cause is **data sparsity**. With only 3 spam emails and a 5-word vocabulary, many word–class combinations never appear in training. The MLE faithfully reports this as probability zero – but zero is catastrophic inside a product, because it destroys all other evidence regardless of magnitude.

This is not a modelling flaw; it is an *estimation* flaw. The true probability that a spam email contains “meeting” is almost certainly not zero – we just have not seen it yet. With a real vocabulary of 50,000 words and a few thousand training emails, the vast majority of word–class combinations will be unseen, making this a severe and pervasive problem.

The fix: Laplace smoothing

We pretend we saw every word at least once in every class by adding a pseudo-count of $\alpha = 1$ to each cell:

$$\hat{\phi}_{j|k}^{(\text{smooth})} = \frac{\sum_{i=1}^N \mathbf{1}[x_j^{(i)} = 1, y^{(i)} = k] + 1}{\sum_{i=1}^N \mathbf{1}[y^{(i)} = k] + 2}.$$

The +1 in the numerator ensures no probability is ever zero. The +2 in the denominator – one for each possible value $x_j \in \{0, 1\}$ – ensures that $P(x_j = 1 \mid y = k)$ and $P(x_j = 0 \mid y = k)$ still sum to 1. The prior $\hat{\phi}_y$ is unchanged.

Laplace-Smoothed Estimates

Prior (unchanged)

$$\hat{\phi}_y = \frac{3}{6} = 0.5$$

Smoothed spam likelihoods ($y = 1$) – denominator: $3 + 2 = 5$

$$\begin{aligned}\hat{\phi}_{\text{free}|1} &= \frac{2+1}{3+2} = \frac{3}{5} = 0.600 \\ \hat{\phi}_{\text{click}|1} &= \frac{2+1}{3+2} = \frac{3}{5} = 0.600 \\ \hat{\phi}_{\text{meeting}|1} &= \frac{0+1}{3+2} = \frac{1}{5} = 0.200 \quad (\text{was } 0.000 \checkmark) \\ \hat{\phi}_{\text{report}|1} &= \frac{0+1}{3+2} = \frac{1}{5} = 0.200 \quad (\text{was } 0.000 \checkmark) \\ \hat{\phi}_{\text{offer}|1} &= \frac{2+1}{3+2} = \frac{3}{5} = 0.600\end{aligned}$$

Smoothed ham likelihoods ($y = 0$) – denominator: $3 + 2 = 5$

$$\begin{aligned}\hat{\phi}_{\text{free}|0} &= \frac{0+1}{3+2} = \frac{1}{5} = 0.200 \quad (\text{was } 0.000 \checkmark) \\ \hat{\phi}_{\text{click}|0} &= \frac{0+1}{3+2} = \frac{1}{5} = 0.200 \quad (\text{was } 0.000 \checkmark) \\ \hat{\phi}_{\text{meeting}|0} &= \frac{2+1}{3+2} = \frac{3}{5} = 0.600 \\ \hat{\phi}_{\text{report}|0} &= \frac{2+1}{3+2} = \frac{3}{5} = 0.600 \\ \hat{\phi}_{\text{offer}|0} &= \frac{0+1}{3+2} = \frac{1}{5} = 0.200 \quad (\text{was } 0.000 \checkmark)\end{aligned}$$

All five previously zero parameters now carry a small but positive value.

Predictions Revisited with Laplace Smoothing

Email $x^{(A)}$: free=1, click=0, meeting=0, report=0, offer=1

Spam score:

$$\begin{aligned} P(x^{(A)} | y = 1) \cdot P(y = 1) \\ &= 0.6 \cdot (1 - 0.6) \cdot (1 - 0.2) \cdot (1 - 0.2) \cdot 0.6 \cdot 0.5 \\ &= 0.6 \times 0.4 \times 0.8 \times 0.8 \times 0.6 \times 0.5 = \mathbf{0.04608} \end{aligned}$$

Ham score:

$$\begin{aligned} P(x^{(A)} | y = 0) \cdot P(y = 0) \\ &= 0.2 \cdot (1 - 0.2) \cdot (1 - 0.6) \cdot (1 - 0.6) \cdot 0.2 \cdot 0.5 \\ &= 0.2 \times 0.8 \times 0.4 \times 0.4 \times 0.2 \times 0.5 = \mathbf{0.00256} \end{aligned}$$

Prediction for $x^{(A)}$ (Laplace)

0.04608 > 0.00256, so we predict $\hat{y} = 1$ (**spam**). ✓ Same outcome as MLE, but now both scores are strictly positive.

Email $x^{(B)}$: free=1, click=0, meeting=1, report=0, offer=1

This was the email that broke the MLE classifier entirely.

Spam score:

$$\begin{aligned} P(x^{(B)} | y = 1) \cdot P(y = 1) \\ &= \hat{\phi}_{\text{free}|1} \cdot (1 - \hat{\phi}_{\text{click}|1}) \cdot \hat{\phi}_{\text{meeting}|1} \cdot (1 - \hat{\phi}_{\text{report}|1}) \cdot \hat{\phi}_{\text{offer}|1} \cdot \hat{\phi}_y \\ &= 0.6 \times 0.4 \times 0.2 \times 0.8 \times 0.6 \times 0.5 = \mathbf{0.01152} \end{aligned}$$

Ham score:

$$\begin{aligned} P(x^{(B)} | y = 0) \cdot P(y = 0) \\ &= \hat{\phi}_{\text{free}|0} \cdot (1 - \hat{\phi}_{\text{click}|0}) \cdot \hat{\phi}_{\text{meeting}|0} \cdot (1 - \hat{\phi}_{\text{report}|0}) \cdot \hat{\phi}_{\text{offer}|0} \cdot (1 - \hat{\phi}_y) \\ &= 0.2 \times 0.8 \times 0.6 \times 0.4 \times 0.2 \times 0.5 = \mathbf{0.00384} \end{aligned}$$

Prediction for $x^{(B)}$ (Laplace)

0.01152 > 0.00384, so we predict $\hat{y} = 1$ (**spam**). ✓

Both scores are strictly positive and the classifier can now make a decision. The presence of **meeting** penalises the spam score through the small factor $\hat{\phi}_{\text{meeting}|1} = 0.2$, but the combined evidence of **free** and **offer** still tips the balance toward spam. Compare to MLE, where both scores were identically zero and no prediction was possible.

Side-by-Side Parameter Comparison

Parameter	MLE formula	MLE value	Laplace formula	Laplace value
$\hat{\phi}_y$	3/6	0.500	3/6	0.500
$\hat{\phi}_{\text{free} 1}$	2/3	0.667	(2+1)/(3+2)	0.600
$\hat{\phi}_{\text{click} 1}$	2/3	0.667	(2+1)/(3+2)	0.600
$\hat{\phi}_{\text{meeting} 1}$	0/3	0.000 ★	(0+1)/(3+2)	0.200
$\hat{\phi}_{\text{report} 1}$	0/3	0.000 ★	(0+1)/(3+2)	0.200
$\hat{\phi}_{\text{offer} 1}$	2/3	0.667	(2+1)/(3+2)	0.600
$\hat{\phi}_{\text{free} 0}$	0/3	0.000 ★	(0+1)/(3+2)	0.200
$\hat{\phi}_{\text{click} 0}$	0/3	0.000 ★	(0+1)/(3+2)	0.200
$\hat{\phi}_{\text{meeting} 0}$	2/3	0.667	(2+1)/(3+2)	0.600
$\hat{\phi}_{\text{report} 0}$	2/3	0.667	(2+1)/(3+2)	0.600
$\hat{\phi}_{\text{offer} 0}$	0/3	0.000 ★	(0+1)/(3+2)	0.200

Key observations

1. $\hat{\phi}_y$ is unchanged by smoothing (no zero-probability issue in the prior).
2. Non-zero MLE estimates shrink slightly toward 0.5, e.g. $\frac{2}{3} \approx 0.667 \rightarrow \frac{3}{5} = 0.600$, as probability mass is redistributed to unseen events.
3. The five zero MLE entries are raised to 0.200, eliminating the zero-product problem entirely.
4. With only 3 training emails per class, smoothing has a pronounced effect. With more data the correction becomes negligible: as $N \rightarrow \infty$, $\hat{\phi}_{j|k}^{(\text{smooth})} \rightarrow \hat{\phi}_{j|k}^{(\text{MLE})}$.